STUDENT-DEVELOPED · IYEC 2026

ROOTED SG

The plant that can be heard.

An intelligent plant-care system that combines plant monitoring, plant-signal sensing and automated care.



Head Pod



Probe in the root zone

SOIL & ENVIRONMENT
Tracks soil conditions and temperature, humidity.

PLANT SIGNALS
Leaf electrode senses electrical differences.

SMART CARE
Decides when to water or add fertiliser.

CONNECTED SYSTEM
ESP32-S3 processes readings; BLE links to the app.

How it works

SENSE → Collect readings.

PROCESS → Apply care logic.

RESPOND → Water or fertiliser when needed.

RECHECK → Inform next care.

BLE → ROOTED SG app



ROOTED SG app

Mojo
AI plant companion.

Plant Doctor
Analyse plant photos and explore problems.

Guardian Path
Interactive learning.



Environmental Impact

- Less water and fertiliser use.
- Support's home and urban gardens.
- More informed plant care.